

# Julia optimization analysis

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***Abstract***—Abstract about libraries comparison.

***Index Terms***—Julia, CNN, optimization, PyTorch, Neural Networks, SDG, Auto Differentiation

## I. INTRODUCTION

This document is a model and instructions for L<sup>A</sup>T<sub>E</sub>X. Please observe the conference page limits.

## II. OPTIMIZATION APPLIED IN OUR LIBRARY

- A. *Types instability*
- B. *im2col*
- C. *float32*
- D. *Weights initialization*
- E. *In-place operations*

## III. COMPARISON WITH OTHER LIBRARIES

- A. *Time complexity*
- B. *Memory usage*

## IV. CONCLUSION

Julia is fast.

## REFERENCES

- [1] G. Eason, B. Noble, and I. N. Sneddon, “On certain integrals of Lipschitz-Hankel type involving products of Bessel functions,” *Phil. Trans. Roy. Soc. London*, vol. A247, pp. 529–551, April 1955.